Evaluating Physics-Informed and Data-Driven Models for Chlorophyll-a Prediction from Landsat 8 Across US Inland Waters

Maya Struzak 12-787 submission

Paper ID *****

Abstract

Chlorophyll-a (chl-a) is a key indicator of harmful algal blooms in inland and coastal waters, yet continuous monitoring remains challenging. Satellite remote sensing offers broad spatial coverage, but chl-a retrieval from Landsat 8 Operational Land Imager (OLI) data is constrained by the absence of red-edge bands and atmospheric correction uncertainties. Machine learning models have shown promise for Landsat-based retrieval, but physics-informed approaches, such as the embedding of bio-optical constraints into the training objective, remain widely unexplored, especially at a national scale. This study evaluates whether incorporating bio-optical constraints into a multilayer perceptron (MLP) loss function improved chl-a prediction from Landsat 8 surface reflectance relative to purely data-driven baselines. Four models were trained and evaluated on a national subset of the AquaSat matchup dataset (n=15,825 spanning diverse US inland and coastal water bodies): a band-ratio linear regression, XGBoost, a plain MLP, and a physics-informed MLP with spectral smoothness and NIR baseline penalties. The XGBoost and both MLP models outperformed the linear regression ($R^2 = 0.13$), achieving $R^2 \approx 0.41$ on the held-out test set. The physics-informed MLP mostly matched but did not exceed the plain MLP, suggesting that the bio-optical constraints are compatible with the data-driven objective but do not provide additional predictive capability at this scale. A regional experiment on the Chesapeake Bay watershed (HUC-4) revealed that sample size is a more critical limiting factor than optical homogeneity for all four models. These results indicate the performance is limited primarily by Landsat 8's spectral range rather than model architecture, and that physics-informed constraints may contribute more meaningfully to sensors with red-edge bands, such as Sentinel-2.

1. Introduction

Chlorophyll-a (chl-a) is the primary green-appearing pigment in photosynthetic organisms, absorbing blue and red light. It is widely used as an indicator of algal biomass in water bodies and is thus useful for detecting and forecasting harmful algal blooms. Agricultural activities can cause excess nutrient loading in water bodies, notably nitrogen and phosphorus species, which lead to explosive algal bloom growth. These algae are considered "harmful" because they create anoxic marine conditions, depleting the available oxygen required for other aquatic organisms, and produce toxins that can contaminate drinking water, recreational areas, and shellfish consumed by humans. Therefore, monitoring algal blooms is important to protect both environmental and human health.

Chl-a concentration is measured primarily through in-situ monitoring and laboratory testing. However, in-situ monitoring data can be spatio-temporally sparse and costly. Satellite imagery serves as a promising alternative, offering broad spatial coverage and relatively consistent data. Specialized band ratio algorithms, such as the Normalized Difference Chlorophyll Index (NDCI) and the Maximum Chlorophyll Index (MCI), are used to estimate chl-a concentrations in coastal and inland waters, leveraging the sensitivity of specific wavelengths to chl-a absorption. [4]. However, they require red-edge bands that are absent from older retrieval products, like Landsat-8 OLI. Landsat-8 retrieval is also more challenging than newer satellite products (i.e. Sentinel-2) due to less accurate atmospheric correction approaches optimized for land surfaces rather than water.

Machine learning models such as random forests, convolutional neural networks, and multilayer perceptrons have shown strong performance for estimating chl-a concentrations from satellite-derived spectral bands in specific bodies of water [1, 9]. XGBoost in particular is a well-supported and competitive model for chl-a retrieval from multispectral remote sensing data, with boosted-tree methods demonstrating promising performance on Landsat-8 OLI [2, 8].

However, few studies have examined physics-informed machine learning approaches for chl-a retrieval, especially at a national scale where new issues arise, including sensor inconsistency and optical heterogeneity across water types. MLPs are especially well-suited for physics-informed extensions, allowing for bio-optical constraints to be directly integrated into the training objective via custom loss functions, which cannot be as naturally integrated into tree-based methods like XGBoost. This study evaluates whether incorporating bio-optical constraints directly into an MLP loss function improves chl-a prediction from Landsat 8 spectral data relative to purely data-driven baselines.

2. Methods

In order to assess the performance of physics-informed MLPs, an open source dataset was used to train, validate, and test three data-driven baseline models and a physics-informed model.

2.1. Dataset

This study uses a subset of *Aquasat* [5], a publicly available dataset pairing Landsat reflectance retrievals with in-situ water quality measurements. Aquasat was constructed by linking field measurements from the Water Quality Portal (WQP) and the LAke multi-scaled GeOSpatial and Temporal database (LAGOS-NE) with the Landsat archive (5, 7, and 8) via Google Earth Engine, yielding over 600,000 match-ups spanning 1984–2019. This study uses the Landsat 8 chl-a subset exclusively, as Landsat 8 is the most recent sensor in the dataset, offering improved radiometric calibration.

Variables. The input features are six Landsat 8 Operational Land Imagery (OLI) surface reflectance bands: blue (∼480nm), green (∼560nm), red (∼655nm), NIR (∼865nm), SWIR1 (∼1610nm), and SWIR2 (∼2200nm). Four band ratios are also computed as input features: green/red, NIR/red, blue/green, and NIR/green. The prediction target is in-situ chl-a concentration (µg/L).

Preprocessing. Band values in AquaSat are stored as scaled reflectance values extracted via Google Earth Engine and were converted to surface reflectance (0–1) by dividing by 10,000, consistent with the GEE extraction pipeline used in AquaSat. Rows with physically impossible reflectance values (outside 0–1) were removed, as well as chl-a values below 0.1 µg/L and above 500 µg/L, corresponding to below-detection-limit and extreme outlier thresholds, respectively. Match-ups with fewer than 9 valid pixels (equivalent to a 3×3 Landsat pixel window) were also excluded to reduce mixed-pixel noise. Additionally, two samples with invalid band ratio values (division by zero due to a zero-valued band) were dropped from the dataset. The chl-a target was log-transformed prior to modeling to correct for strong right skew; the log-transformed distribution is approximately normal (QQ plot R^2=0.998, Figure 1).

Train/validation/test split. After pre-processing, 15,825 samples remained, spanning 2013–2019 across inland and coastal water bodies within the contiguous US (Figure 2). These were split into training, validation, and test sets (with respectivley, n=12,660 or 80%, n = 1,582 or 10%, and n = 1,583 or 10%) using stratified random splitting on log(chl-a) decile bins to preserve the target distribution when partitioning. All input features were normalized to zero mean and unit variance using a StandardScaler fit on the training set, only, which were then applied to the validation and test sets to prevent data leakage. A random seed of 42 was used for data splitting, model initialization, and training.

2.2. Data-Driven Models

Three data-driven baseline models are used to contextualize the performance of a physics-informed MLP: a plain multi-layer perceptron (MLP), an XGBoost regressor, and a simple linear regression on a single band ratio.

2.2.1. Plain MLP

The MLP has 10 input features and a single output neuron predicting log(chl-a). The network consists of three hidden layers with a narrowing width of 128 $\rightarrow$ 64 $\rightarrow$ 32, each followed by batch normalization, ReLU activation, and dropout (rate=0.2). The model is trained using mean squared error (MSE) loss on log(chl-a) with the Adam optimizer (learning rate =0.001) and batch size of 256. Early stopping with a patience of 20 epochs is applied to prevent overfitting.

2.2.2. XGBoost

The XGBoost regressor uses 500 boosting rounds with a maximum tree depth of 6, a learning rate of 0.05, and subsampling and column sampling rates of 0.8. Instead of a gradient descent optimizer, XGBoost uses a tree-based boosting optimizer. Early stopping was applied using the validation set.

2.2.3. Linear Regression

The linear regression model fits a least-squares regression between the blue/green band ratio and log(chl-a). This ratio was selected as it showed the strongest linear correlation with log(chl-a) among all spectral features in the training set (Pearson r=-0.35), and matches traditional ocean color band-ratio algorithms.

2.3. Physics-Informed MLP

The physics-informed model extends the baseline MLP by incorporating constraints informed by bio-optical relationships between surface reflectance and chl-a concentration into the standard MSE loss function, maintaining an architecture identical to the plain MLP (except for the loss

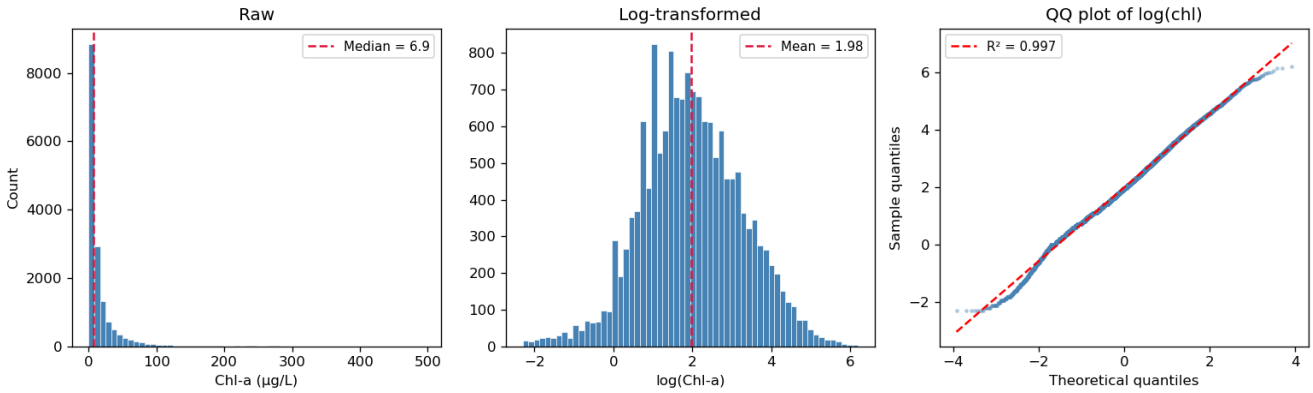

Figure 1. Chl-a distribution of raw (left) and log-transformed (center) values. QQ-plot (right) confirms normality of log-transformed distribution.

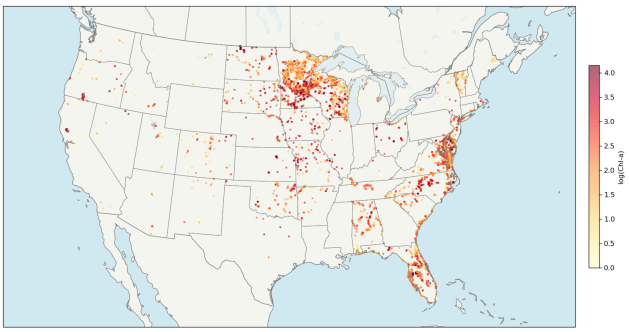

Figure 2. Geographic distribution of Landsat 8 chl-a matchups (n=15,825)

function). Additionally, the physical constraint of non-negativity is already enforced implicitly in the data preprocessing for all models: chl-a concentrations are predicted in the log-space, as the exponential transformation produces only positive outputs, and negative surface reflectance band values were excluded from the dataset.

The constraints incorporated into the baseline include the following:

1. ***Spectral smoothness:*** real water reflectance spectra are physically smooth, without abrupt discontinuities between bands. The spectral smoothness penalty is applied to the four visible and near-infrared bands (blue, green, red, NIR), which span a contiguous spectral range. The SWIR bands are excluded because they are separated by a large wavelength gap from the visible bands, making a smoothness penalty physically inappropriate. Thus, this penalty discourages large differences between adjacent visible and NIR band values for a given sample (Eq. 1).

$$\mathcal{L}_{\text{smooth}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{B-1}(r_{i,j+1} - r_{i,j})^2 \qquad (1)$$

where $r_{i,j}$ denotes the reflectance of the j-th band for sample i, with $j = 1, 2, 3, 4$ corresponding to blue ($\sim$480nm), green ($\sim$560nm), red ($\sim$655nm), and NIR ($\sim$865nm) respectively, and $B = 4$.

2. ***NIR baseline:*** Since water absorbs most NIR light, NIR reflectance should be near zero over open water. Anomalously high NIR reflectance indicates likely contamination from suspended sediment or mixed land-water pixels, producing physically unreliable input signals. This constraint penalizes samples where NIR reflectance exceeds a data-driven threshold (Eq. 2).

$$\mathcal{L}_{\text{NIR}} = \frac{1}{N}\sum_{i=1}^{N}\max(0,\ r_{i,\text{NIR}} - \tau)^2 \qquad (2)$$

where $r_{i,\text{NIR}}$ is the NIR reflectance for sample i and τ is a threshold set at the 90th percentile of NIR reflectance in the training set.

Both constraints operate on the input reflectance values passed to the model, requiring no additional learned parameters. The penalty terms are computed during each forward pass and added to the MSE loss before backpropagation. These penalties act as input-space regularizers, penalizing physically implausible reflectance signals during training rather than constraining model outputs directly. The hyperparameters λ_1 and λ_2 are tuned on the validation set, starting at zero (equivalent to the plain MLP baseline) on a grid search over 0, 0.001, 0.01, 0.1, with optimal values selected based on validation RMSE ($\lambda = 0.01$ for both). The combined loss function is thus:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_1\mathcal{L}_{\text{smooth}} + \lambda_2\mathcal{L}_{\text{NIR}} \qquad (3)$$

where λ_1 and λ_2 are hyperparameters controlling the influence of each physics penalty.

2.4. Regional Experiment

To evaluate whether a regionally-constrained model outperforms the national model a specific watershed, a geographic subset corresponding to the Chesapeake Bay watershed was extracted from the AquaSat Landsat 8 dataset [3]. The Chesapeake Bay watershed was selected for its status as the largest HUC-4 watershed, as well as its ecological, economic, and social significance. The regional data was split into training, validation, and test sets (with respectively, n=1,798 or 80%, n=255 or 10%, and n=255 or 10%), using the same stratified splitting procedure and random seed (42) as the national experiment. All four models were retrained from scratch on this subset using identical hyperparameters to the national models; no hyperparameter retuning was performed.

2.5. Evaluation

All four models are evaluated on the held-out test set using three metrics: RMSE in log-space, R^2 in log-space and RMSE in original units ($\mu g/L$). The use of log-space units reflects model training objective, whereas original units allow interpretability of real-world error magnitude. Qualitative analyses include predicted vs. observed scatter plots (Figure 4).

3. Results

Test performance is reported below for all four models at both the national and regional scales, followed by qualitative analysis of model predictions. All metrics are reported on held-out test data not seen during training or hyperparameter tuning.

3.1. National-scale model comparison

Table 1 reports the performance of the four models in the test set of the national AquaSat Landsat 8 dataset (n = 1,583 test samples).

The band-ratio regression achieved R^2 = 0.13, confirming that a single spectral index is insufficient for national-scale chl-a retrieval. All three ML models substantially outperformed the linear regression, achieving $R^2 \approx$ 0.40–0.41 and RMSE $\approx$ 0.99–1.00 in log-space. XGBoost marginally outperformed both MLP variants for RMSE (=0.99), but matched the plain MLP and physics-informed MLP with $R^2 = 0.41$. Both MLPs had RMSE = 1.00 in log-space. In original concentration units, the physics-informed MLP showed marginally higher RMSE than the plain MLP (26.65 vs. 26.51 µg/L), suggesting the physics penalties did not improve — and slightly degraded — performance in concentration space

Training and validation loss curves for the plain and physics-informed MLPs (Figure 3) showed smooth convergence with no divergence between train and val loss, indicating the absence of overfitting. Early stopping triggered at epoch 82, with the best validation loss at approximately epoch 62. Prediction versus actual plots (Figure 4) show reasonable agreement along the 1:1 line in log-space, with increasing scatter at high chl-a concentrations — a known challenge in inland water remote sensing due to the optical complexity of bloom conditions.

3.2. Regional experiment: Chesapeake Bay watershed

All models performed worse on the HUC-4 Chesapeake Bay watershed subset (Table 2) than on the national dataset (Table 1), with XGBoost achieving the highest R^2 = 0.32 and both MLP variants achieving R^2 = 0.19. The degraded MLP performance on the regional subset suggests that sample size is a more important limiting factor than optical heterogeneity at this scale. The regional training set (n = 1,798) appears insufficient for stable MLP convergence, whereas XGBoost, which typically requires less data, degraded more slowly. Similar to the national-scale analyses, the plain MLP has a slightly higher RMSE in original units than the physics-informed MLP, despite having identical R^2 and RMSE scores in the log-space.

4. Discussion

All three ML model performances converge on $R^2 \approx 0.40$–0.41 on the national test set, which is indicative of limitations imposed by Landsat 8's spectral configuration rather than model architecture. Unlike Sentinel-2, which provides red-edge bands at 705 and 740 nm that are directly sensitive to chl-a fluorescence and absorption, Landsat 8 OLI lacks these wavelengths. The strongest single predictor identified in our analysis–the blue/green ratio (Pearson $r = -0.35$)–is substantially weaker than the MCI and NDCI indices available with Sentinel-2, which have been shown to achieve $r > 0.5$ in comparable datasets [6]. This spectral limitation is consistent with prior work showing that Landsat-8 chl-a retrieval is fundamentally constrained by the absence of red-edge bands and atmospheric correction uncertainties [2, 7].

The physics-informed MLP matched but did not exceed the plain MLP baseline. The bio-optical constraints–spectral smoothness across visible and NIR bands, and a NIR baseline penalty–did not provide additional predictive signal beyond what the data-driven objective captured. This result suggests that at the scale of 12,660 training samples drawn from diverse US water bodies, the MLP has sufficient capacity to learn spectrally consistent representations without explicit regularization. The physics constraints did not degrade performance, indicating compatibility with the MSE training objective, and may provide greater benefit in low-data settings or when applied to out-of-distribution water bodies not well-represented in AquaSat. It is also

Table 1. Test set performance for all models on the national AquaSat Landsat 8 dataset. Metrics are reported in log-space and original concentration units ($\mu g/L$).

Model	R^2 (log)	RMSE (log)	RMSE ($\mu g/L$)
Band-ratio regression	0.13	1.20	30.33
XGBoost	0.41	0.99	25.97
Plain MLP	0.41	1.00	26.51
Physics-informed MLP	0.41	1.00	26.65

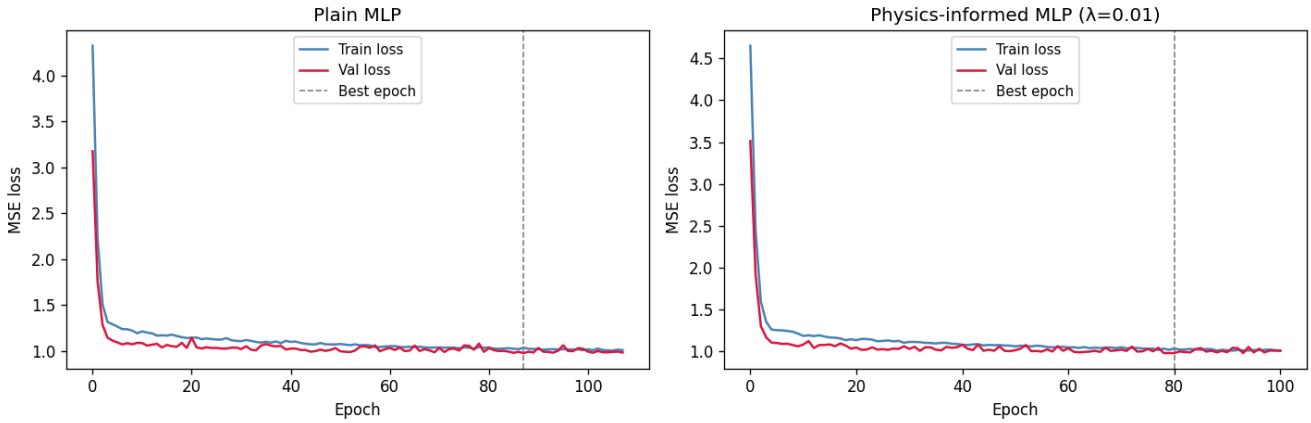

Figure 3. Training and validation loss curves for the Plain MLP (left) and Physics-Informed MLP on the national dataset

important to acknowledge the constraints were applied to pre-cleaned inputs, which may have reduced the benefit of penalties, especially for NIR. Furthermore, the chosen constraints may be insufficiently targeted for Landsat 8's spectral configuration, which lacks the red-edge bands most directly linked to chl-a bio-optical relationships, limiting the potential benefit of bio-optical regularization.

The regional Chesapeake Bay experiment revealed that performance degraded when the national model was replaced by a regionally-trained one, contradicting the expectation that optical homogeneity would improve predictions. This finding suggests that for MLP-based models, training set size is a more important factor than optical consistency at this spatial scale. XGBoost was more data-efficient in the regional setting, maintaining $R^2 = 0.32$ with only 1,798 training samples compared to $R^2 = 0.19$ for both MLP variants — consistent with known advantages of ensemble tree methods in low-sample regimes.

4.1. Limitations.

1. The dataset is geographically biased toward the Upper Midwest and Southeast US, reflecting the distribution of WQP monitoring stations rather than the distribution of US water bodies. Model performance in sparsely sampled regions, particularly the western US, may be lower than reported test set metrics suggest.
2. Although AquaSat match-ups with fewer than 9 valid pixels were excluded, the 3×3 pixel window (90m) may still capture mixed land-water signals in narrow or complex water bodies, potentially introducing noise in training and evaluation.
3. The bio-optical penalties operate on input reflectance values rather than constraining model outputs directly. As a result, they function as input-space regularizers and do not guarantee physically plausible predictions for out-of-distribution inputs.

5. Conclusion

This study evaluated whether incorporating bio-optical constraints into a physics-informed MLP loss function improves chl-a prediction from Landsat 8 surface reflectance data at national scale, using the AquaSat matchup dataset of 15,825 observations across diverse US inland and coastal water bodies. Four models were compared: a band-ratio linear regression, XGBoost, a plain MLP, and a physics-informed MLP with spectral smoothness and NIR baseline penalties.

All three ML models substantially outperformed the linear regression baseline (R^2 = 0.13), achieving $R^2 \approx 0.41$ on the test set. The physics-informed MLP mostly matched the performance of both XGBoost and the plain MLP, suggesting that bio-optical constraints are compatible with the data-driven training objective but do not provide additional predictive capability at this scale. In log-space, the physics-

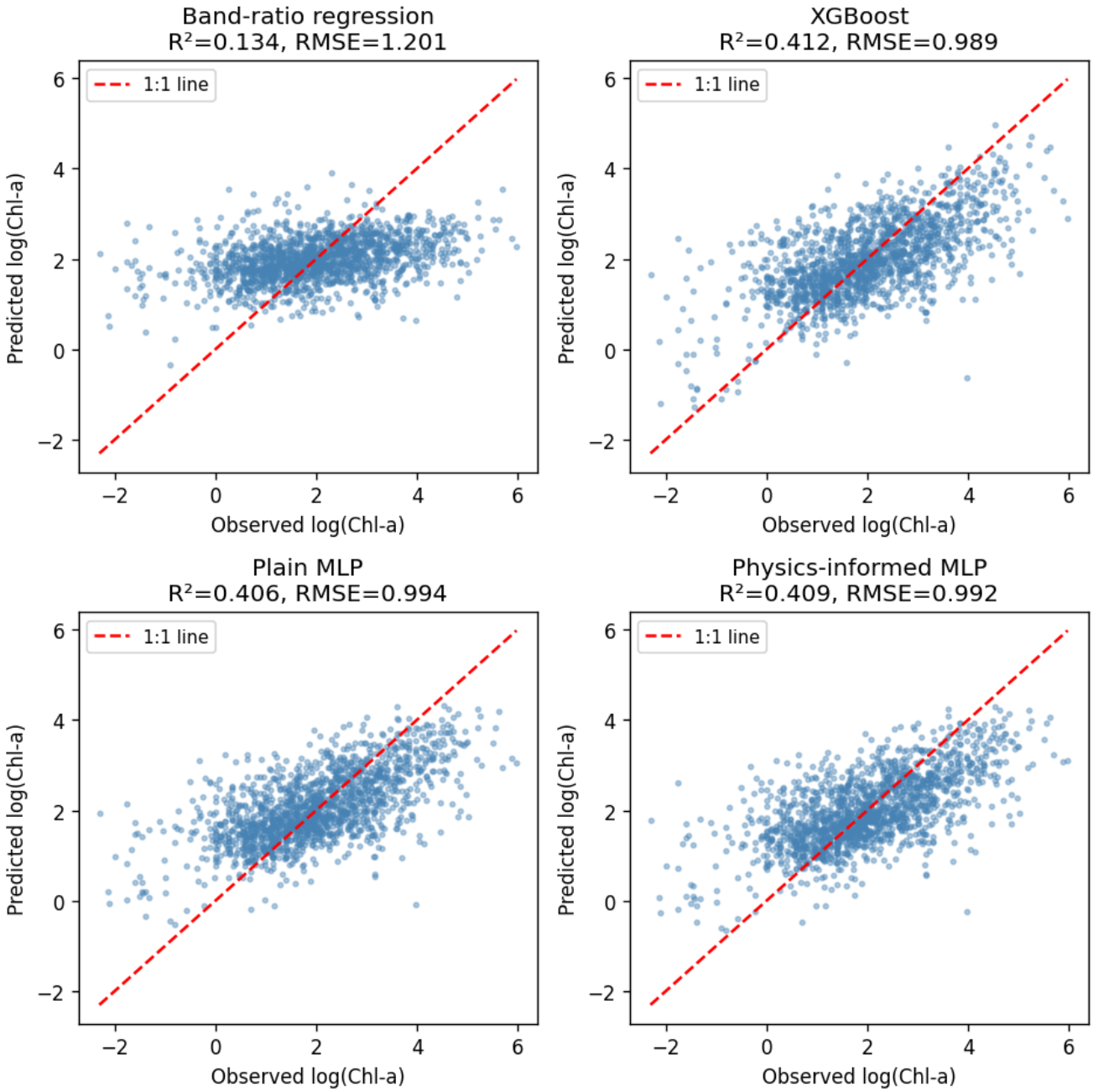

Figure 4. Predicted vs observed log(chl-a) values based on test set of national dataset for all four models

Table 2. Test set performance for all models on Chesapeake Bay watershed subset of AquaSat Landsat 8

Model	R^2 (log)	RMSE (log)	RMSE (µg/L)
Band-ratio regression	0.01	1.16	21.44
XGBoost	0.32	0.96	19.05
Plain MLP	0.19	1.05	19.98
Physics-informed MLP	0.19	1.05	19.76

informed MLP was marginally best, and in original units it was marginally worse. The performance ceiling appears to be driven by Landsat 8's limited spectral resolution for chl-a retrieval (specifically the absence of red-edge bands at 705 and 740 nm) rather than model architecture. A regional experiment on the Chesapeake Bay watershed further demonstrated that sample size is a more critical limiting factor than optical homogeneity for MLP-based models at sub-national scales.

These findings suggest multiple directions for future work. First, applying the physics-informed MLP framework to Sentinel-2 data, which provides red-edge bands more directly sensitive to chl-a, may allow the physics constraints to contribute more meaningfully and push performance beyond the Landsat-imposed ceiling. Second, incorporating additional bio-optical constraints, such as the expected co-variation between chl-a and NIR reflectance in productive waters, or exploring water-type-specific models may improve performance in optically complex settings. The physics-informed architecture developed here provides a flexible framework for integrating domain knowledge into water quality retrieval, with potential applications beyond chl-a to other optically-detectable constituents such as total suspended solids and dissolved organic carbon.

6. Code Availability

Code is available at https://github.com/mstruzak/chla-landsat8-piml.

References

[1] Vicky Anand, Bakimchandra Oinam, and Silke Wieprecht. Machine learning approach for water quality predictions based on multispectral satellite imageries. *Ecological Informatics*, 84:102868, 2024. 1

[2] Zhigang Cao, Ronghua Ma, Hongtao Duan, Nima Pahlevan, John Melack, Ming Shen, and Kun Xue. A machine learning approach to estimate chlorophyll-a from Landsat-8 measurements in inland lakes. *Remote Sensing of Environment*, 248: 111974, 2020. 1, 4

[3] US Fish and Wildlife Service. Chesapeake Bay Watershed Boundary, 2023. 4

[4] Sachidananda Mishra and Deepak R. Mishra. Normalized difference chlorophyll index: A novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sensing of Environment*, 117:394–406, 2012. 1

[5] Matthew Ross. AquaSat, 2019. 2

[6] Pahlevan N Coffer MM Seegers BN Werdell PJ Ferriby H Stumpf RP Binding CE Keith DJ Salls WB, Schaeffer BA. Expanding the application of sentinel-2 chlorophyll monitoring across united states lakes. *Remote Sens (Basel)*, 16(11):1-29, 2024. 4

[7] Brandon Smith, Nima Pahlevan, John Schalles, Steve Ruberg, Reagan Errera, Ronghua Ma, Claudia Giardino, Mariano Bresciani, Claudio Barbosa, Tim Moore, Virginia Fernandez, Krista Alikas, and Kersti Kangro. A Chlorophyll-a Algorithm for Landsat-8 Based on Mixture Density Networks. *Frontiers in Remote Sensing*, 1, 2021. 4

[8] Eden T. Wasehun, Leila Hashemi Beni, Courtney A. Di Vittorio, Christopher M. Zarzar, and Kyana R.L. Young. Comparative analysis of Sentinel-2 and PlanetScope imagery for chlorophyll-a prediction using machine learning models. *Ecological Informatics*, 85:102988, 2025. 1

[9] Zhengxiao Yan and Nasrin Alamdari. Projecting harmful algal blooms in North Biscayne Bay using machine learning and bias-corrected climate scenarios. *Journal of Hydrology: Regional Studies*, 62:102865, 2025. 1